

Master IARFID

Statistical Structured Prediction, Part II: Fundamentals of Multimodal Interaction in Pattern Recognition

Enrique Vidal

evidal@prhlt.upv.es



Pattern Recognition and Human Language Technology Research Center

Departamento de Sistemas Informáticos y Computación

Universitat Politècnica de València

January 2023

E. Vidal – PRHLT-UPV-DSIC

Master IARFID: Predicción Estructurada Estadística (PEE) Statistical Structured Prediction

- Part I (Joan Andreu Sánchez & José Miguel Benedí)
Statistical Structured Prediction for structured output data
- Part II (Enrique Vidal)
Fundamentals of Multimodal Interaction in Pattern Recognition
 1. Pattern Recognition (PR) and Person-Machine Interaction
 2. Feedback, Multimodality and Adaptive Learning in IPR
 3. User Models, Interaction Protocols and Assessment
 4. Interaction-driven learning
 5. Applications

E. Vidal – PRHLT-UPV-DSIC

Index

Introduction

- 1 Notation and introduction to Interactive Pattern Recognition ▷ 0

Classical Pattern Recognition (PR)

- 2 PR and structured-output prediction ▷ 8
- 3 A running example: Kariotype recognition ▷ 12

Interactive PR (IPR): Feedback and multimodal processing

- 4 Directly benefit from human feedback ▷ 20
- 5 Non-deterministic feedback and multimodal IPR ▷ 31

User Models, Interaction Protocols and Assessment

- 6 Passive, active and other interaction protocols ▷ 41
- 7 Estimating user interaction effort ▷ 49

Interaction-driven learning

- 8 Adaptive, on-line, active and reinforcement learning ▷ 54

Applications, Final Remarks and Bibliography

- 9 Applications ▷ 61
- 10 Future work and Conclusions ▷ 65
- 11 Bibliography ▷ 68

E. Vidal – PRHLT-UPV-DSIC

IARFID – PEE

II MIPR theory

Notation and Basic Concepts in Statistics

- UNCONDITIONAL, CONDITIONAL AND JOINT PROBABILITIES:

$$\Pr(X = x), \quad \Pr(X = x \mid Y = y), \quad \Pr(X = x, Y = y)$$

$$\text{Notation: } P(x), \quad P(x \mid y), \quad P(x, y)$$

$$(\text{also: } P_{\mathcal{M}}(x) \equiv P(x), \quad P_{\mathcal{M}}(x \mid y) \equiv P(x \mid y), \quad \text{etc.})$$

- BAYES' RULE: $P(x, y) = P(x) \cdot P(y \mid x) = P(y) \cdot P(x \mid y)$

- CHAIN RULE:

$$P(x_1, x_2, \dots, x_n) = P(x_1) \cdot P(x_2 \mid x_1) \cdots P(x_n \mid x_1, \dots, x_{n-1})$$

$$\text{Notation (for sequences): } P(x_1^n) = P(x_1) \cdot P(x_2 \mid x_1) \cdots P(x_n \mid x_1^{n-1})$$

$$\text{Naive Bayes approximation: } P(x_1^n) \approx P(x_1) \cdot P(x_2) \cdots P(x_n)$$

- MARGINAL: $P(x) = \sum_y P(x, y)$

- **MODE**: $\hat{x} = \arg \max_x P(x)$; $P(\hat{x}) = \max_x P(x)$

- **MODE** APPROXIMATION: $\sum_x P(x, \dots) \approx \max_x P(x, \dots)$

Interactive, Computer-assisted Pattern Recognition: Motivation

- In most Pattern Recognition (PR) problems and applications, development purportedly aims at fully automated systems
- But full automation often proves elusive or unnatural in many applications where technology is expected to *assist*, rather than replace the human agents
- In these and many other cases, *practical* PR developments typically end up just in “semiautomatic systems” or systems for “*computer assisted*” operation, where it is a human expert who makes the final decisions
- The traditional *Training-Test partition* PR paradigm proves inadequate in many applications of increasing interest. Manual work is needed *both* to annotate the training data and to fix system errors in the test phase
- These facts are very seldom acknowledged: typically, full automation is pretended and the “eventual” need of human intervention is ignored in the mathematical formulation (it is often left as an “implementation detail”)

Computer assistance and/or human interaction require a paradigm shift in PR which entails interesting research challenges and opportunities

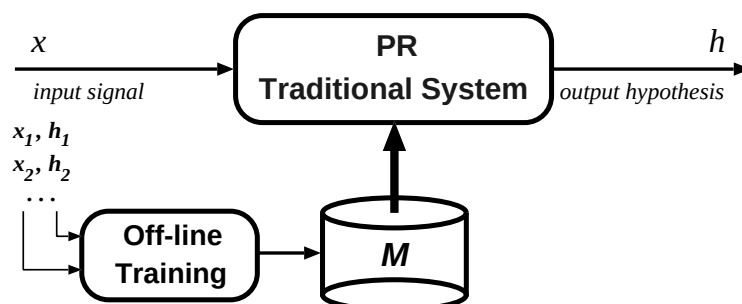
Interactive Pattern Recognition (IPR): Challenges & Opportunities

Opportunities and challenges entailed by *human interaction* in PR:

1. *Feedback information* directly derived from the interaction process can be used to significantly *improve system performance*
2. Feedback signals are generally of a nature or *modality* different from that of the main signals of the original PR problem. *Multimodal synergy* helps to *improve overall system behavior and usability*
3. Each interaction step yields valuable ground-truth data. This promotes *adaptive training* as a means to *tune system performance* for the specific task and/or user mode of operation

Traditional Pattern Recognition

Full Automation

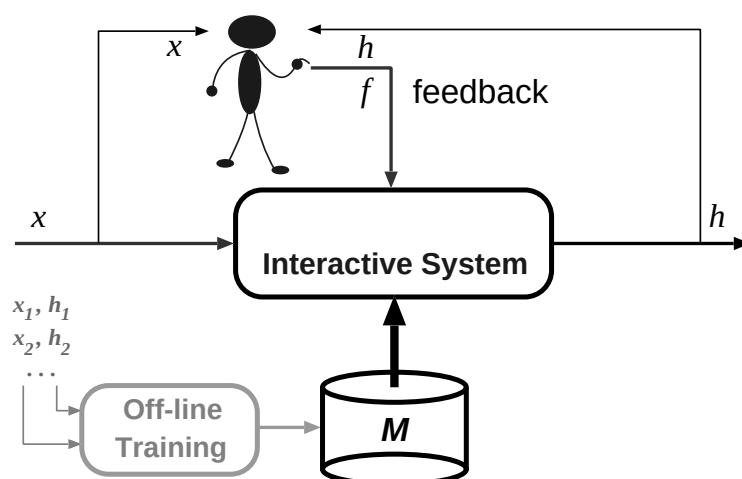


A best hypothesis is one which maximizes the posterior probability, approximated by models $\mathcal{M}$ “batch-trained” from *training pairs* (x_i, h_i) :

$$\hat{h} = \arg \max_h P(h | x) \approx \arg \max_h P_{\mathcal{M}}(h | x) = \arg \max_h P_{\mathcal{M}}(x | h) P_{\mathcal{M}}(h)$$

Interactive Pattern Recognition

Human Feedback

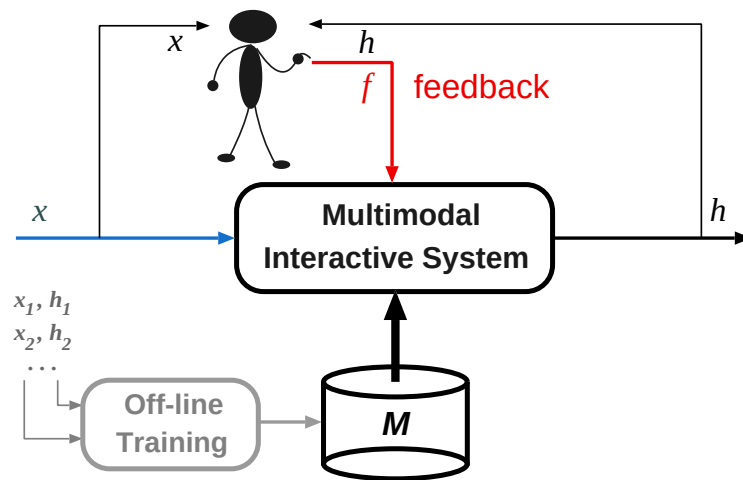


Interaction feedback entails adding more *conditions*, which allow improving system output hypotheses:

$$\hat{h} \approx \arg \max_h P_{\mathcal{M}}(h | x, f) \approx \arg \max_h P_{\mathcal{M}}(x | h) P_{\mathcal{M}}(h | f)$$

Multimodal Interactive Pattern Recognition

Multimodality

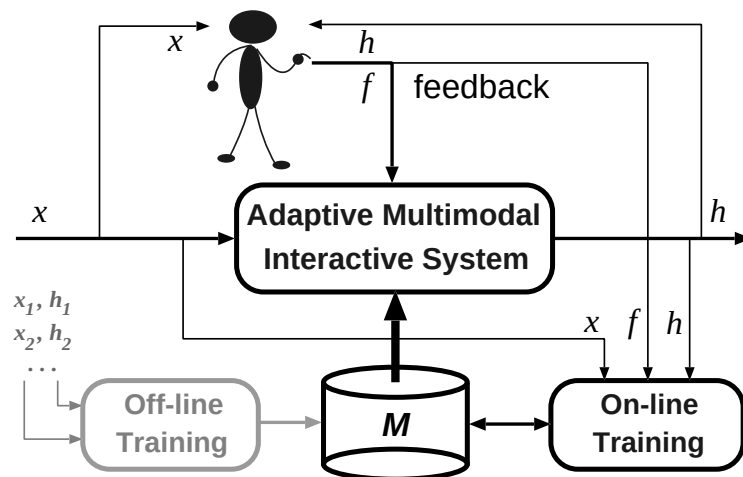


Main (x) and feedback (f) signals seldom belong to the same domain; hence IPR naturally entails an intrinsic form of *Multimodal Processing*:

$$\hat{h} \approx \arg \max_h P_{\mathcal{M}}(h | x, f) \approx \arg \max_h P_{\mathcal{M}_X}(x | h) \cdot P_{\mathcal{M}_F}(f | h) \cdot P_{\mathcal{M}_H}(h)$$

Adaptive Multimodal Interactive Pattern Recognition

Adaptivity



Feedback data allows to **adaptively (re-)train** $\mathcal{M}$ and tune the system to changing environment

Index

Introduction

- 1 Notation and introduction to Interactive Pattern Recognition ▷ 0

Classical Pattern Recognition (PR)

- 2 *PR and structured-output prediction* ▷ 8
- 3 A running example: Karyotype recognition ▷ 12

Interactive PR (IPR): Feedback and multimodal processing

- 4 Directly benefit from human feedback ▷ 20
5 Non-deterministic feedback and multimodal IPR ▷ 31

User Models, Interaction Protocols and Assessment

- 6 Passive, active and other interaction protocols ▷ 41
7 Estimating user interaction effort ▷ 49

Interaction-driven learning

- 8 Adaptive, on-line, active and reinforcement learning ▷ 54

Applications, Final Remarks and Bibliography

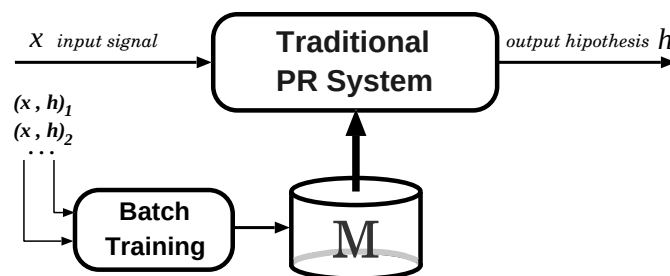
- 9 Applications ▷ 61
10 Future work and Conclusions ▷ 65
11 Bibliography ▷ 68

Classical Pattern Recognition and Decision Theory

- Decision theory is adopted to minimize the cost of wrong hypotheses.
- In the simplest case, a 0/1 cost function is used which corresponds to minimizing the *number* of wrong hypotheses (*minimal error criterion*).
- Under the *minimal error criterion*, a best hypothesis is shown to be one which maximises the posterior probability $P(h | x)$. Using a model $\mathcal{M}$, this is approximated as:

$$\hat{h} = \arg \max_{h \in \mathcal{H}} P(h | x) \approx \arg \max_{h \in \mathcal{H}} P_{\mathcal{M}}(h | x) \quad (1)$$

where $\mathcal{H}$ is the (possibly infinite) set of valid hypotheses.



Classical Pattern Recognition and model training

- Minimal error is also the main Decision Theory criterion adopted for development of *statistical learning* approaches to train $\mathcal{M}$ from the training data (an example of these approaches is *Maximum likelihood*)
- However, in many cases it is difficult to directly estimate $P_{\mathcal{M}}(h \mid x)$ and it is better to apply the Bayes rule to decompose Eq. (1) as:

$$\hat{h} \approx \arg \max_{h \in \mathcal{H}} P(h \mid x) = \arg \max_{h \in \mathcal{H}} P(x \mid h) \cdot P(h) \quad (2)$$

Two models need to be estimated:

- The *likelihood* model(s) $P(x \mid h)$, which can often be easily estimated from the available training pairs (x_i, h_i) , $i = 1, 2, \dots$
- The *prior* $P(h)$, which can be easily estimated by using only the output data, h_i , of the available training pairs.

Both estimates can be carried out following the maximum likelihood approach.

Classification and structured-output prediction

- **Classification:** most traditional and simple PR framework where $\mathcal{H} = \{1, \dots, C\}$ and, typically, C is small
 - Useful PR framework, with many applications
 - Only *trivial, exhaustive search* is needed to solve $\arg \max_{h \in \mathcal{H}} P(c \mid x)$
- **Structured-output prediction:** $\mathcal{H}$ is a possibly infinite space where each $h \in \mathcal{H}$ is *structured* into a *sequence, graph, set*, etc. of *hypothesis elements*
 - Applications of increasing interest: *Automatic Speech or Handwritten Text Recognition* (ASR, HTR), *Machine Translation* (MT), *Image and Video Processing*, etc.; outputs (h) are sequences of words, arrays of labels, etc.
 - Both *hypothesis search* and *model training* may become very complex; but several search and training approaches exist: Viterbi search, probabilistic relaxation, belief propagation, Baum-Welch estimation, etc.
 - IPR can be particularly useful; for instance, in HTR, human feedback can consist of signaling and/or to fixing elementary errors (such as a misrecognized word or character), rather than full transcripts.

Index

Introduction

- 1 Notation and introduction to Interactive Pattern Recognition ▷ 0

Classical Pattern Recognition (PR)

- 2 PR and structured-output prediction ▷ 8

- 3 *A running example: Karyotype recognition* ▷ 12

Interactive PR (IPR): Feedback and multimodal processing

- 4 Directly benefit from human feedback ▷ 20

- 5 Non-deterministic feedback and multimodal IPR ▷ 31

User Models, Interaction Protocols and Assessment

- 6 Passive, active and other interaction protocols ▷ 41

- 7 Estimating user interaction effort ▷ 49

Interaction-driven learning

- 8 Adaptive, on-line, active and reinforcement learning ▷ 54

Applications, Final Remarks and Bibliography

- 9 Applications ▷ 61

- 10 Future work and Conclusions ▷ 65

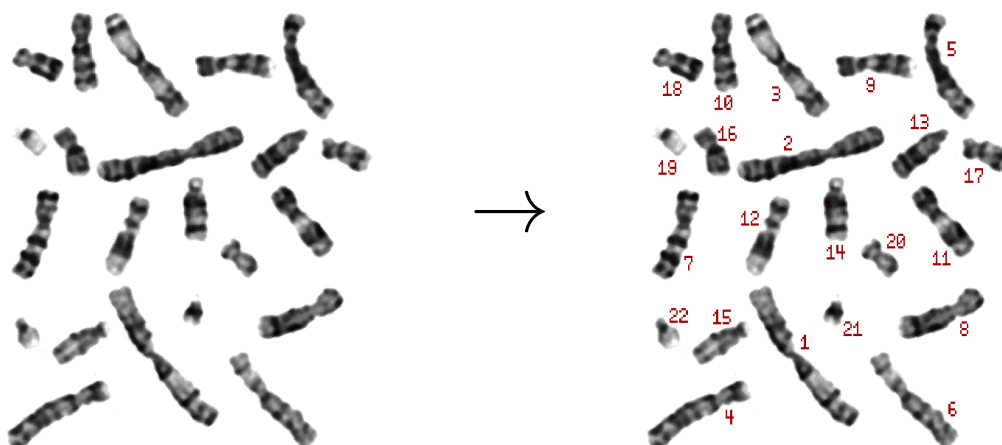
- 11 Bibliography ▷ 68

Example: Recognition of human karyotypes

Given a set of 46 unsorted images of stained human chromosomes, label each image from a set of 24 labels, $\{1, 2, \dots, 22, X, Y\}$, in such a way that each label is assigned exactly to two images (except for X, Y , where assignments are sex-dependent).

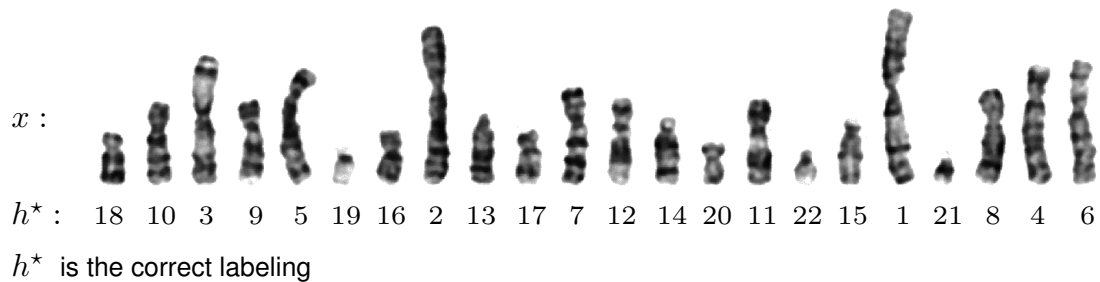
Simplification: (consider only single images, rather than pairs, and ignore chromosomes X, Y):

Given a set of 22 *unsorted images* of stained human chromosomes, label each image from a set of 22 *labels*, $\{1, 2, \dots, 22\}$, in such a way that each label is assigned exactly to *one* image.



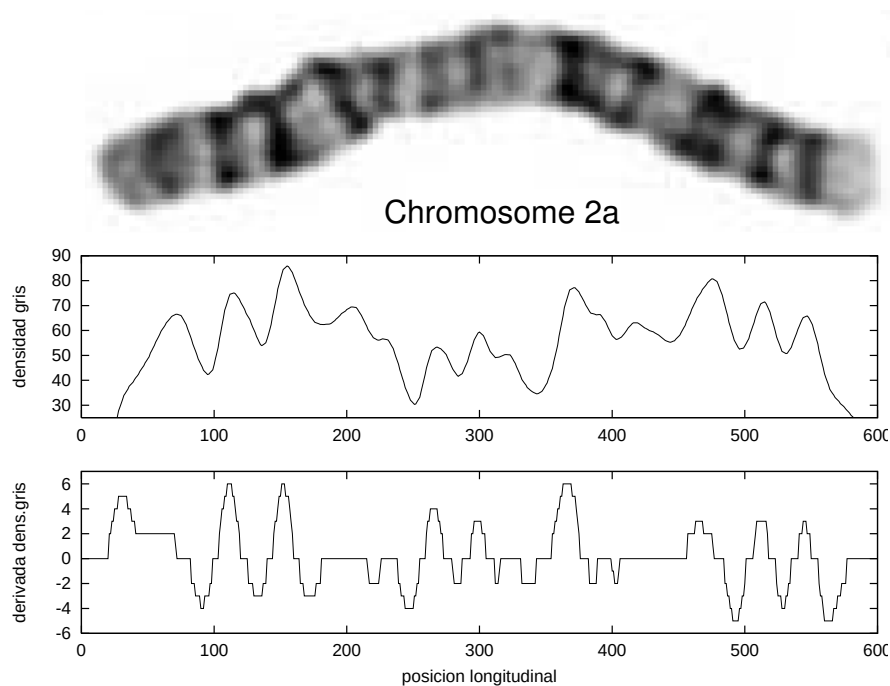
Human karyotyping: representation and notation

- $x = x_1, \dots, x_{22} \equiv x_1^{22} \in \mathcal{X}$ is an unsorted sequence of 22 chromosome images, arranged from left to right in some arbitrary order.
- $h = h_1^{22} \in \mathcal{H}$ is a sequence of 22 labels, $h_i \in \{“1”, “2”, \dots “22”\}, 1 \leq i \leq 22$
- $\mathcal{H}$ is finite but huge ($|\mathcal{H}| = 22^{22}$)



- Each individual chromosome image x_i is represented, for example, as a grey-level projection profile of the chromosome image on its median axis

Chromosome image representation



Example: Classical PR of individual chromosomes

In the vast majority of works carried out so far each chromosome image, x_i is recognized with independence of the others:

$$\hat{h}_i = \arg \max_{c \in \{“1”, \dots, “22”\}} P(c | x_i) = \arg \max_{c \in \{“1”, \dots, “22”\}} P(x_i | c) P(c) \quad 1 \leq i \leq 22$$

- *Prior*: All the 22 chromosome classes, are equiprobable; i.e., $P(c) = 1/22$
- *Likelihood*: $P(x_i | c)$ can be approximated, for instance, by a hidden Markov model for each chromosome class, $P(x_i | c)$

Therefore:

$$\hat{h}_i = \arg \max_{c \in \{“1”, \dots, “22”\}} P(x_i | c) \quad 1 \leq i \leq 22$$

Example: Classical PR of individual chromosomes

In the vast majority of works carried out so far each chromosome image, x_i is recognized with independence of the others:

$$\hat{h}_i = \arg \max_{c \in \{“1”, \dots, “22”\}} P(c | x_i) = \arg \max_{c \in \{“1”, \dots, “22”\}} P(x_i | c) P(c) \quad 1 \leq i \leq 22$$

- *Prior*: All the 22 chromosome classes, are equiprobable; i.e., $P(c) = 1/22$
- *Likelihood*: $P(x_i | c)$ can be approximated, for instance, by a hidden Markov model for each chromosome class, $P(x_i | c)$

Therefore:

$$\hat{h}_i = \arg \max_{c \in \{“1”, \dots, “22”\}} P(x_i | c) \quad 1 \leq i \leq 22$$

Problem:

$\hat{h}$ may not be a karyotype labelling (it may have repeated and missing labels).

Example: Classical PR of individual chromosomes

In the vast majority of works carried out so far each chromosome image, x_i is recognized with independence of the others:

$$\hat{h}_i = \arg \max_{c \in \{“1”, \dots, “22”\}} P(c | x_i) = \arg \max_{c \in \{“1”, \dots, “22”\}} P(x_i | c) P(c) \quad 1 \leq i \leq 22$$

- *Prior*: All the 22 chromosome classes, are equiprobable; i.e., $P(c) = 1/22$
- *Likelihood*: $P(x_i | c)$ can be approximated, for instance, by a hidden Markov model for each chromosome class, $P(x_i | c)$

Therefore:

$$\hat{h}_i = \arg \max_{c \in \{“1”, \dots, “22”\}} P(x_i | c) \quad 1 \leq i \leq 22$$

Problem:

$\hat{h}$ may not be a karyotype labelling (it may have repeated and missing labels).

Another (more insidious) problem:

Without karyotype restriction, individual classification errors are high

Example: classical PR of karyotypes

$$\hat{h} \approx \arg \max_{h \in \mathcal{H}} P(x | h) P(h) \quad (3)$$

- $P(h)$: full-labelling prior probability; $P(x | h)$: image sequence likelihood
- The *prior* is well known, but not trivial. Ideally, it should be *null* if h contains repeated symbols and *flat* otherwise; that is:

$$P(h) = \begin{cases} \frac{1}{22!} & \text{if } h_i \neq h_j \forall i \neq j, 1 \leq i, j \leq 22 \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

- The *likelihood* can be approached by a naive Bayes decomposition:

$$P(x | h) = P(x_1, \dots, x_{22} | h_1, \dots, h_{22}) \approx \prod_{i=1}^{22} P(x_i | h_i) \quad (5)$$

- As in the case of individual chromosomes, each $P(x_i | h_i)$ can be modeled by a hidden Markov model

- But, how to solve (3)?

Example: search for classical PR of karyotypes

- An exact solution to the search problem (3) is difficult because of the huge size of $\mathcal{H}$ and the tangled restrictions entailed by $P(h)$ (no repeated labels)
- But a simple *greedy* approximation can provide acceptable results:
 - First, for each individual chromosome image, x_j , compute its max-posterior, $\max_{c \in \{“1”, \dots, “22”\}} P(c \mid x_j)$ (similar as for individual chromosome image classification)
 - Sort the images according to these scores
 - Then, following this max-posterior order, assign to each chromosome image, x_i , the label $\hat{h}_i = \arg \max_{k \in \mathcal{K}} P(k \mid x_i)$, taking care that labels assigned to previous images can no longer be assigned; that is $\mathcal{K} = \{“1”, \dots, “22”\} - \{\hat{h}_1, \dots, \hat{h}_{i-1}\}$
- Obviously, this can only achieve local optimisation, since other complete labellings $h \neq \hat{h}$ may exist for which $P(h \mid x) > P(\hat{h} \mid x)$
- Optimal search solutions by means of *Branch & Bound* or *Dynamic Programming* methods

Example: classical PR karyotyping results

- Experiments with the so-called “Copenhagen Chromosomes Data Set”: 200 karyotypes and 4 400 chromosome samples
- Split into two blocks of 100 karyotypes (2 200 chromosome samples)
- Two-block Cross-Validation; results averaged over the two runs

Karyotype and chromosome error rate (in %)

Approach	Chromosome	Karyotype
Individual chromosomes	8.0	76
Using prior $P(h)$ (greedy)	3.7	27
Using prior $P(h)$ (B & B)	2.2	15

[Oncina & Vidal, 2011]

Index

Introduction

- 1 Notation and introduction to Interactive Pattern Recognition ▷ 0

Classical Pattern Recognition (PR)

- 2 PR and structured-output prediction ▷ 8
3 A running example: Karyotype recognition ▷ 12

Interactive PR (IPR): Feedback and multimodal processing

- 4 *Directly benefit from human feedback* ▷ 20
5 Non-deterministic feedback and multimodal IPR ▷ 31

User Models, Interaction Protocols and Assessment

- 6 Passive, active and other interaction protocols ▷ 41
7 Estimating user interaction effort ▷ 49

Interaction-driven learning

- 8 Adaptive, on-line, active and reinforcement learning ▷ 54

Applications, Final Remarks and Bibliography

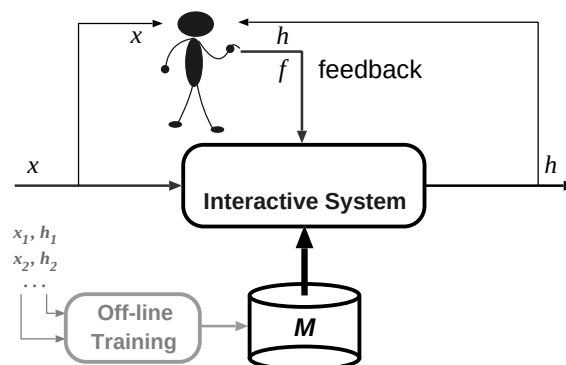
- 9 Applications ▷ 61
10 Future work and Conclusions ▷ 65
11 Bibliography ▷ 68

Interactive Pattern Recognition and Multimodal Interaction

Feedback: Take direct advantage of the *feedback information* provided by the user in each interaction step to improve raw performance,

Multimodality: It arises as a *natural property of interaction*. By properly acknowledging this fact, improved overall system performance and usability can be achieved.

Adaptation: Use feedback-derived data to *adaptively* (re-)train the system and tune it to the user behaviour and the specific task considered.



Using the Human Feedback Directly

- In classical PR, for a fixed model $\mathcal{M}$, and given x , a best hypothesis, $\hat{h}$, is one which maximises the posterior probability $P_{\mathcal{M}}(h \mid x)$.
- In IPR, without varying $\mathcal{M}$, $\hat{h}$ can be improved by adding more *conditions*:

$$\hat{h} = \arg \max_{h \in \mathcal{H}} P_{\mathcal{M}}(h \mid x, f) \quad (6)$$

$f \in \mathcal{F}$ represents the feedback, interaction-derived informations;
e.g., in the form of *partial hypothesis* or *constraints* on $\mathcal{H}$.

- The new system hypothesis, $\hat{h}$, may prompt the user to provide further feedback, thereby starting a new interaction step.
- The process continues until the system output is acceptable by the user.
- The richer the feedback informations, f , the better $\hat{h}$ can be obtained
- But modelling and search for (6) may be (much) more difficult than with our familiar $P_{\mathcal{M}}(h \mid x)$.

Factorizing the Feedback-conditioned Posterior

- The prediction of $\hat{h}$ can be conveniently developed by factorizing $P_{\mathcal{M}}(h \mid x, f)$:

$$\hat{h} = \arg \max_{h \in \mathcal{H}} P(h \mid x, f) = \arg \max_{h \in \mathcal{H}} P(x \mid f, h) P(h \mid f)$$

- $P(x \mid f, h)$, can be considered independent[†] of f given h ; so:

$$\hat{h} = \arg \max_{h \in \mathcal{H}} P(x \mid h) P(h \mid f)$$

Similar to classical PR, but now the *prior* is feedback conditioned

User feedback often refers to previous statuses of the interactive process, making it convenient to explicitly consider *interaction history*.

[†]Details discussed later.

Explicitly Taking Interaction History into Account

- History from previous interaction steps can be easily taken into account
- The history, h' , can be represented by the optimal hypothesis, $\hat{h}$, obtained by the system in its previous interaction step[†] for the given x
- Since previous hypotheses have been supervised/corrected by the user, a part of h' is correct for the given x . In the current interaction step, the feedback f aims at further correcting element(s) of h' . Taking history into account, Eq. (6) becomes:

$$\hat{h} = \arg \max_{h \in \mathcal{H}} P(h \mid x, h', f) \quad (7)$$

Algorithm IPR–History // Let x be the input and $\hat{h}$ the output hypothesis
 $\hat{h} = \arg \max_{h \in \mathcal{H}} P(h \mid x)$ // Initialization
do forever { // Interaction loop
 $f = \text{user_feedback}(\hat{h})$; **if** ($f = \text{“OK”}$) **return** $\hat{h}$
 $h' = \hat{h}$; $\hat{h} = \arg \max_{h \in \mathcal{H}} P(h \mid x, h', f)$
}

[†]This is a first-order approach. More generally, h' can represent an adequate combination of the optimal hypotheses obtained in all previous interaction steps for the given x

Interaction with Deterministic Feedback

- In general, feedback signals have to be recognized or decoded. Let $\mathcal{D}$ be the space of decoded feedback signals.
- *Deterministic* feedback modalities (e.g., keyboard & mouse), greatly simplify matters. Feedback decoding can then be specified as a function, $d : \mathcal{F} \rightarrow \mathcal{D}$, which maps each raw feedback signal, f , into its corresponding (trivial and unique) decoding $d = d(f)$.

For instance, if f is the signal of a keystroke on the key “A”, $d(f)$ is the symbol “A” itself (keyboards are not expected to produce erroneous output symbols!).

- The feedback, f can be replaced by its decoding, d ; therefore:

$$\hat{h} = \arg \max_{h \in \mathcal{H}} P(h \mid x, h', d) = \arg \max_{h \in \mathcal{H}} P(x \mid h', d, h) P(h \mid h', d) \quad (8)$$

- $P(x \mid h', d, h)$, can be considered independent of h' and d given h (in fact d typically conveys information aimed to modify an element or a part of h'); so:

$$\hat{h} = \arg \max_{h \in \mathcal{H}} P(x \mid h) P(h \mid h', d) \quad (9)$$

Similar to classical PR, but now the *prior* is history and feedback conditioned

Interaction with Deterministic Feedback (2)

- The pair (h', d) can be seen as a partially amended version of h' , where one or more errors from the last step have been corrected. So, typically:

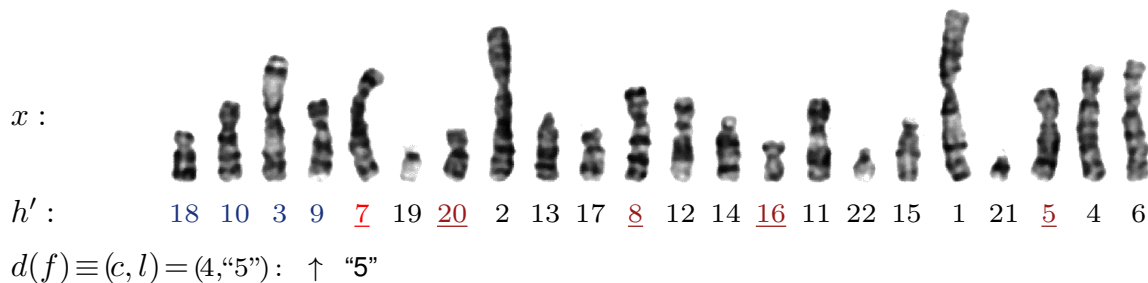
$$P(h \mid h', d) \propto \begin{cases} P(h) & \text{if } h \text{ is compatible with } (h', d) \\ 0 & \text{otherwise} \end{cases} \quad (10)$$

- These model changes can be interpreted just as a part of the search problem by substituting $\mathcal{H}$ with a smaller space, $\mathcal{H}' \subset \mathcal{H}$, in which the feedback-derived restrictions apply.
- This way, an IPR problem can often be seen as a variation of the corresponding non-interactive PR problem where identical models are used but the search strategy has to be changed:

$$\hat{h} = \arg \max_{h \in \mathcal{H}'} P(x \mid h) P(h) \quad (11)$$

Example: Interactive Karyotyping

- In non-interactive karyotype recognition *each* individual chromosome label error had to be manually amended or “*post-edited*”
- In the interactive framework, the system may take advantage of each manual correction to improve its hypotheses for the remaining chromosome images.
- Clearly, this may significantly reduce the amount of human effort needed to produce a correct karyotype:



Example of keyboard & pointer interaction in simplified human karyotyping. Labeling errors are marked in red and underlined. User feedback consists in positioning the cursor over the last correct label ($c = 4$, $h_c = \text{"9"}$) and then typing the correction ($l = \text{"5"}$) on the next position.

Correct labels: $h^* = 18, 10, 3, 9, 5, 19, 16, 2, 13, 17, 7, 12, 14, 20, 11, 22, 15, 1, 21, 8, 4, 6$

Example: Interactive Karyotyping: IPR formulation

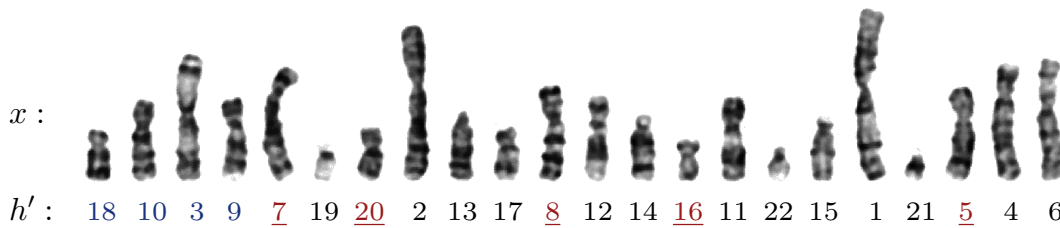
- An initial karyotype $\hat{h}$ is obtained using Eqs. (3-5). In each successive interaction step, $\hat{h}$ becomes the history, h' , and a new $\hat{h}$ is obtained using (9).
- In each step, the user feedback, $f \in \mathcal{F}$, consists of keystrokes to specify a position c in h' where the last correct label appears, and a label $l \in \{“1”, “2”, \dots “22”\}$ to fix the first labelling error.
- Since f is *deterministic*, it is trivially “decoded” as $d = d(f) \equiv (c, l) \in \mathcal{D}$. The first wrong label in h' is h'_{c+1} and its correct value should be l .
- These interaction-derived informations condition the possible values of h as:

$$\begin{aligned} h_1^c &= h_1'^c \\ h_{c+1} &= l \\ h_i &\notin \{h'_1, \dots, h'_c, l\}, \quad c+2 \leq i \leq 22 \end{aligned} \quad (12)$$

- Let $\mathcal{H}'(h', c, l)$ be the subset of hypotheses, h , that comply (12). The *conditioned prior* can be written as:

$$P(h | h', d) = P(h | h', c, l) = \begin{cases} \propto P(h) \text{ [as in (4)]} & \text{if } h \in \mathcal{H}'(h', c, l) \\ 0 & \text{otherwise} \end{cases} \quad (13)$$

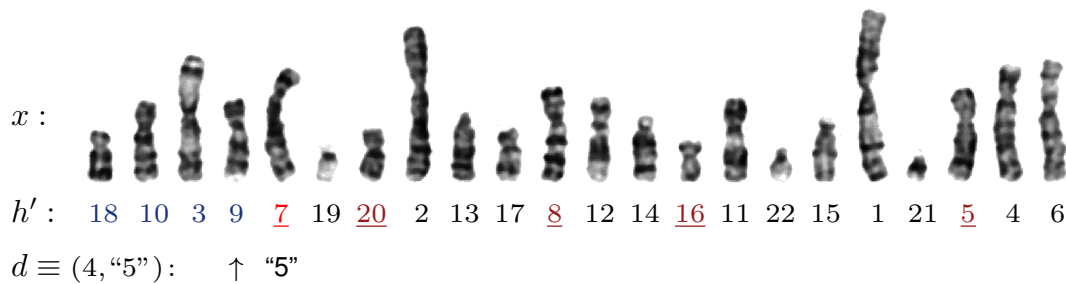
Example: Interactive Karyotyping



Example of keyboard & pointer interaction in simplified human karyotyping. A first result is obtained by solving the classical (non-interactive) karyotype recognition problem. The resulting hypothesis is a *valid* karyotype, but has 5 labeling errors, marked in red and underlined.

Correct labels: $h^* = 18, 10, 3, 9, 5, 19, 16, 2, 13, 17, 7, 12, 14, 20, 11, 22, 15, 1, 21, 8, 4, 6$

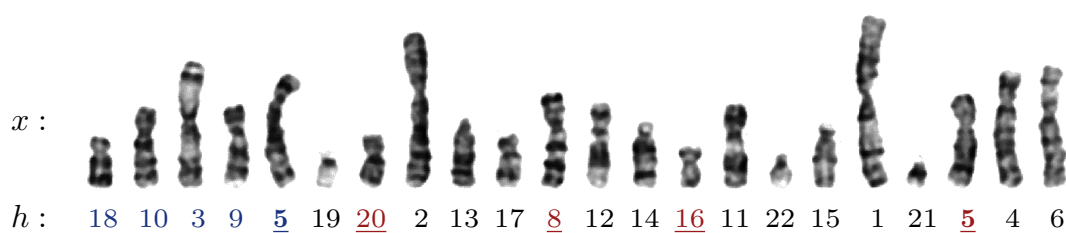
Example: Interactive Karyotyping: first correction feedback



Example of keyboard & pointer interaction in simplified human karyotyping. Labeling errors are marked in red and underlined. User feedback consists in positioning the cursor over the last correct label ($c = 4$, $h_c = "9"$) and then typing the correction ($l = "5"$) on the next position.

Correct labels: $h^* = 18, 10, 3, 9, 5, 19, 16, 2, 13, 17, 7, 12, 14, 20, 11, 22, 15, 1, 21, 8, 4, 6$

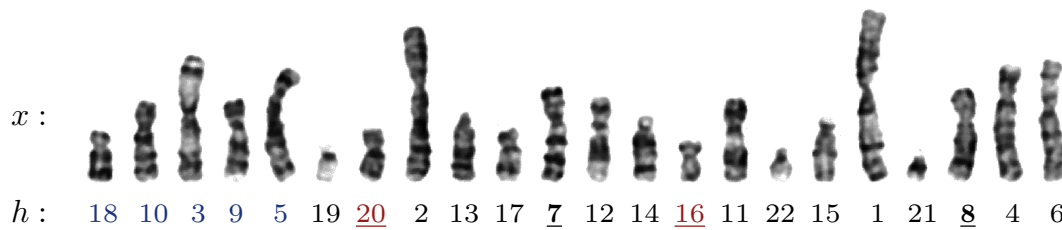
Example: Interactive Karyotyping: apply corrective feedback



Example of keyboard & pointer interaction in simplified human karyotyping. Labeling errors are marked in red and underlined. Now h is not a valid karyotype (the label "5" is repeated and "7" is missing).

Correct labels: $h^* = 18, 10, 3, 9, 5, 19, 16, 2, 13, 17, 7, 12, 14, 20, 11, 22, 15, 1, 21, 8, 4, 6$

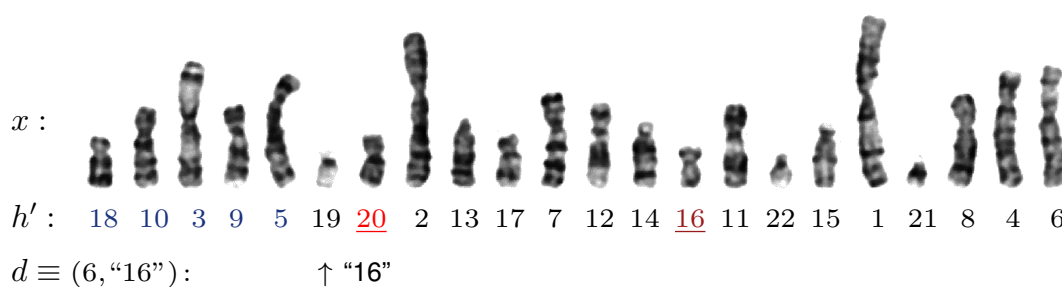
Example: Interactive Karyotyping: first prediction



Example of keyboard & pointer interaction in simplified human karyotyping. Labeling errors are marked in red and underlined. Two previously wrong labels ("8" and "5") have automatically been fixed (into "7" and "8", respectively) thanks to the preceding correction feedback.

Correct labels: $h^* = 18, 10, 3, 9, 5, 19, 16, 2, 13, 17, 7, 12, 14, 20, 11, 22, 15, 1, 21, 8, 4, 6$

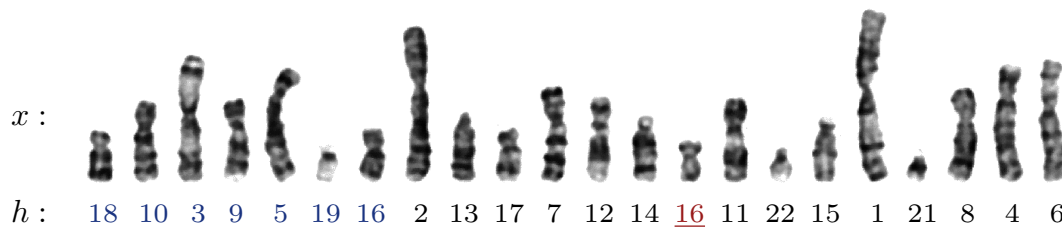
Example: Interactive Karyotyping: new correction feedback



Example of keyboard & pointer interaction in simplified human karyotyping. Labeling errors are marked in red and underlined. User feedback consists in positioning the cursor over the last correct label ($c = 6$, $h_c = "19"$) and then typing the correction ($l = "16"$) on the next position.

Correct labels: $h^* = 18, 10, 3, 9, 5, 19, 16, 2, 13, 17, 7, 12, 14, 20, 11, 22, 15, 1, 21, 8, 4, 6$

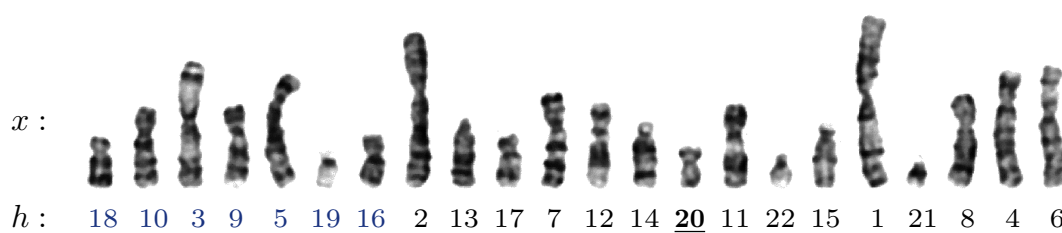
Example: Interactive Karyotyping: apply corrective feedback



Example of keyboard & pointer interaction in simplified human karyotyping. Labeling errors are marked in red and underlined. Now h is not a valid karyotype (the label “16” is repeated and “20” is missing).

Correct labels: $h^* = 18, 10, 3, 9, 5, 19, 16, 2, 13, 17, 7, 12, 14, 20, 11, 22, 15, 1, 21, 8, 4, 6$

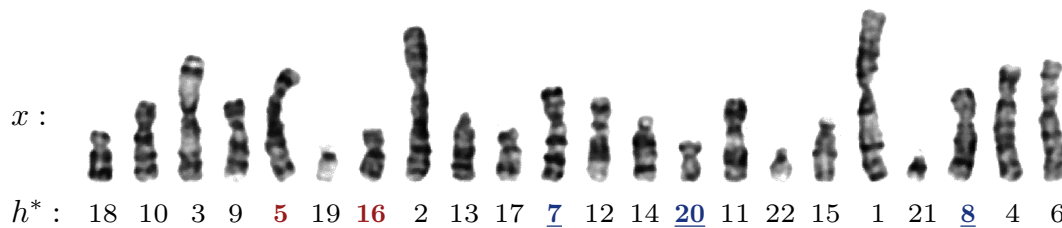
Example: Interactive Karyotyping: second and last prediction



Example of keyboard & pointer interaction in simplified human karyotyping. The remaining label error (“20”) has been automatically corrected (into “16”) thanks to the preceding corrective feedback.

Correct labels: $h^* = 18, 10, 3, 9, 5, 19, 16, 2, 13, 17, 7, 12, 14, 20, 11, 22, 15, 1, 21, 8, 4, 6$

Example: Interactive Karyotyping: Performance



Example of keyboard & pointer interaction in simplified human karyotyping. The initial karyotype had 5 label errors; 2 of them (in boldface red) have been manually fixed by the user and the interactive system has automatically corrected the remaining 3 (in boldface blue and underlined).

Example: Interactive Karyotyping results

- Same experimental conditions as for classical PR karyotyping results
- Interactive (IPR) results using both greedy and B & B search

Number of corrections needed (%)

Approach	Chromosome	Karyotype
Individual Chromosomes	8.0	76
Using prior $P(h)$ (greedy)	3.7	27
Using prior $P(h)$ (B & B)	2.2	15
IPR (greedy)	2.1	27
IPR (B & B)	1.1	15

[Oncina & Vidal, 2011]

Index

Introduction

- 1 Notation and introduction to Interactive Pattern Recognition ▷ 0

Classical Pattern Recognition (PR)

- 2 PR and structured-output prediction ▷ 8
- 3 A running example: Kariotype recognition ▷ 12

Interactive PR (IPR): Feedback and multimodal processing

- 4 Directly benefit from human feedback ▷ 20
- 5 *Non-deterministic feedback and multimodal IPR* ▷ 31

User Models, Interaction Protocols and Assessment

- 6 Passive, active and other interaction protocols ▷ 41
- 7 Estimating user interaction effort ▷ 49

Interaction-driven learning

- 8 Adaptive, on-line, active and reinforcement learning ▷ 54

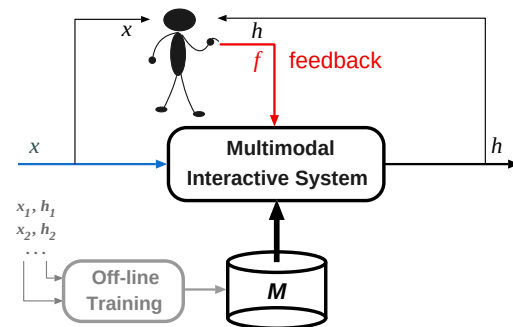
Applications, Final Remarks and Bibliography

- 9 Applications ▷ 61
- 10 Future work and Conclusions ▷ 65
- 11 Bibliography ▷ 68

Multimodal Interaction

In general, feedback informations, $f \in \mathcal{F}$, do not naturally belong to the original domain from which the main data, x , come from; i.e., $\mathcal{F} \neq \mathcal{X}$.

For instance, in a vehicle plate recognition system, it is quite unlikely that user's feedback comes in the form of images obtained by the same camera used to capture the plate images. Instead, it will arrive in form of keystrokes, mouse gestures, or perhaps spoken utterances



- If feedback is non-deterministic, interaction entails some sort of *multimodality* (in addition to the possible multimodal nature of the input signal(s))
- Multimodality appears in many areas of Computer Science and Engineering. The challenge here is how to achieve an adequate *modality synergy* which finally allows taking maximum advantage from all the modalities involved, as well as from the underlying interaction-derived constraints.

Basic Multimodal Fusion

- *Modality fusion*: given two signals, u , v , of some multimodal datum, z , find a best hypothesis, $\hat{h}$, about z ; that is:

$$\hat{h} = \arg \max_{h \in \mathcal{H}} P_{\mathcal{M}}(h \mid u, v) = \arg \max_{h \in \mathcal{H}} P_{\mathcal{M}}(u, v \mid h) \cdot P_{\mathcal{M}}(h) \quad (14)$$

Here, conditional independence of u and v given h can often be assumed

- For instance, in an image description or labelling problem, let u be an image and v the signal of a spoken utterance about the image. Thanks to the independence assumption:

$$\hat{h} = \arg \max_{h \in \mathcal{H}} P_{\mathcal{M}_U}(u \mid h) \cdot P_{\mathcal{M}_V}(v \mid h) \cdot P_{\mathcal{M}_H}(h) \quad (15)$$

- This simple *naïve Bayes* decomposition allows a separate estimation of independent models, $\mathcal{M}_U$, $\mathcal{M}_V$ and $\mathcal{M}_H$, for the image and speech components, and the labelling language, respectively
- The only “joint” problem here is the joint optimisation in (15). This approximation is often known as “*late fusion*”

In IPR, u corresponds to the input signal, x , and v to the feedback, f .

Using Interaction Information to Help Decoding Non-Deterministic Feedback Signals

In IPR, the modalities u , v are the input and the feedback and (15) becomes:

$$\hat{h} \approx \arg \max_{h \in \mathcal{H}} P(x \mid h) \cdot P(f \mid h) \cdot P(h) \quad (16)$$

Here the decoding of f , d , is “*hidden*”. Actual decoding of the feedback is not really needed to obtain $\hat{h}$, but it may be useful for several reasons, including adaptive learning (discussed later). From, (7), it can be uncovered as follows:

$$\hat{h} = \arg \max_h P(h \mid x, h', f) = \arg \max_h \sum_d P(h, d \mid x, h', f) \quad (17)$$

Approximating the sum with the mode, applying basic probability rules and ignoring terms which do not depend on the optimisation variables (h and d):

$$\hat{h} \approx \arg \max_h \max_d P(h \mid h', d, x, f) \cdot P(d \mid h', x) \cdot P(f \mid d, h', x) \quad (18)$$

Then, using the Bayes rule as needed, and making various independence assumptions:

$$(\hat{h}, \hat{d}) \approx \arg \max_{h, d} P(f \mid d) \cdot P(d \mid h') \cdot P(x \mid h) \cdot P(h \mid h', d) \quad (19)$$

Using Interaction Information to Help Decoding Non-Deterministic Feedback Signals: Modelling

$$(19) \rightarrow (\hat{h}, \hat{d}) \approx \arg \max_{h,d} P(f | d) \cdot P(d | h') \cdot P(x | h) \cdot P(h | h', d)$$

- The last two terms of this equation are the same used in (9) for the basic IPR formulation with deterministic feedback
- The other two terms now deal with the non-deterministic feedback:
 - $P(f | d)$ is a *feedback likelihood* model, as in conventional PR for recognizing f
 - $P(d | h')$ is a history-conditioned *feedback decoding prior*,
- Except for the history condition on the prior, these are the terms that would be needed in Eq. (2) for conventional recognition of feedback signals
- But now the conditioned prior is more informative and, moreover, Eq. (19) entails a joint optimisation for simultaneous recognition of main (x) and feedback (f) data
- Clearly, this offers opportunities for more accurate feedback decoding than just using a conventional, off-the-shelf PR system for feedback signals recognition

Using Interaction Information to Help Decoding Non-Deterministic Feedback Signals: Search

$$(19) \rightarrow (\hat{h}, \hat{d}) \approx \arg \max_{h,d} P(f | d) \cdot P(d | h') \cdot P(x | h) \cdot P(h | h', d)$$

This joint optimisation is difficult and it seldom admits exact and efficient solutions. But there are simple and adequate approximations.

Simplest idea: decompose (19) into a two-phase computation:

1. Obtain an “optimal” feedback decoding, $\hat{d}$, using the available history, but ignoring informations directly related with the main data, (x):

$$\hat{d} = \arg \max_d P(f | d) \cdot P(d | h') \quad (20)$$

2. Using the fixed $\hat{d}$, the first two terms of the optimisation (19) become independent of both d and h , which leads to Eq. (21), identical to (9).

$$\hat{h} \approx \arg \max_h P(x | h) \cdot P(h | h', \hat{d}) \quad (21)$$

Using Interaction Information to Help Decoding Non-Deterministic Feedback Signals: Search (1)

$$(19) \rightarrow (\hat{h}, \hat{d}) \approx \arg \max_{h,d} P(f | d) \cdot P(d | h') \cdot P(x | h) \cdot P(h | h', d)$$

The previous simple idea can be easily improved to take into account information from the main data.

1. In the first phase, rather than computing just an “optimal” $\hat{d}$, obtain a list of the n most probable decodings:

$$\{\hat{d}_1, \dots, \hat{d}_n\} = \underset{d}{n\text{-best}} P(f | d) \cdot P(d | h') \quad (22)$$

2. Apply n times the same techniques used to solve (21) or (9) to solve:

$$\hat{h} \approx \arg \max_h \max_{1 \leq i \leq n} P(f | \hat{d}_i) \cdot P(\hat{d}_i | h') \cdot P(x | h) \cdot P(h | h', \hat{d}_i) \quad (23)$$

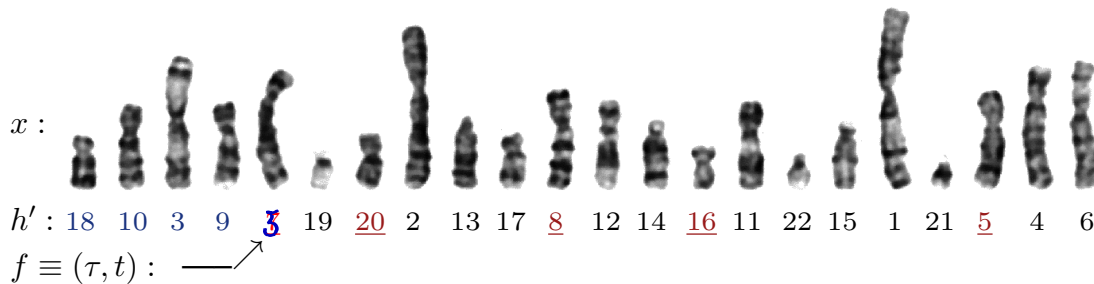
As a byproduct, an optimal $\hat{d}_i$ is obtained, which is possibly better than the one given by the basic “two-phase” approach (20).

Karyotyping example: non-Deterministic feedback

Example of feedback provided by an *e-pen* interface: f is a *sequence of points* or *trajectory* of the pen tip, which encompasses two different parts:

- A deterministic one, τ , consisting in the first point of f , which unambiguously determines the position, $c + 1$, of the first wrong label in h'
- A *non-deterministic* trajectory of amending pen-strokes, t , corresponding to the remaining points of f , which has to be *decoded* into an optimal label, $\hat{l}$

That is, for a feedback signal $f \equiv (\tau, t)$, its decoding will be $d \equiv (c, \hat{l})$.



Errors are in red and underlined. The e-pen corrective feedback, f , is in blue: a digit “5”, handwritten over the first wrong label, “3”. The possible decodings of f would be pairs (c, l) such as $(4, “3”)$, $(4, “5”)$, $(4, “6”)$, . . . , hopefully including the correct decoding, $(4, “5”)$.

Non-Deterministic feedback karyotyping example: models & search

$$(19) \rightarrow (\hat{h}, \hat{d}) \approx \arg \max_{h, d} P(f | d) \cdot P(d | h') \cdot P(x | h) \cdot P(h | h', d)$$

- *Feedback decoding likelihood models:*

$$P(f | d) \equiv P(\tau, t | c, l) = P(\tau | c, l)P(t | c, l, \tau) = P(t | l)$$

It can be modelled with HMMs, as in conventional *on-line* HTR

- *History-conditioned chromosome label prior:*

$$P(d | h') \equiv P(c, l | h') = P(c | h')P(l | h', c) = P(l | h', c)$$

It should be null for already validated labels in $h'_1{}^c$ and for the wrong h'_{c+1} ; flat for the other labels. *Note:* without interaction-derived information, the best prior would be just a uniform distribution over $\{“1”, “2”, \dots “22”\}$,

- The other two models are as in the deterministic-feedback case: $P(h | h', c, l)$ must be null for all h with repeated symbols and for those h such that $h_1^{c+1} \neq h'_1, \dots, h'_c, l$, or $h_i \in \{h_1, \dots, h_c, l\}$, $c + 2 \leq i \leq 22$; flat otherwise.

Search: To solve (19), both search solutions (20–21) and (22–23) can be used, along with the greedy search of the conventional, non-interactive case.

Non-Deterministic Feedback Signals: concluding remarks

- Non-deterministic feedback decoding will never be error-free
- With respect to using deterministic feedback, non-deterministic multimodal interfaces will always increase the number of interaction steps needed to accomplish a given task.

In other words, some degree of performance has to be sacrificed for a potentially improved ergonomy and/or user friendliness

- The design of a good non-deterministic multimodal feedback interface ultimately amounts to achieving a maximum feedback decoding accuracy by taking the maximum possible advantage of contextual informations provided by the interactive framework.

Index

Introduction

- 1 Notation and introduction to Interactive Pattern Recognition ▷ 0

Classical Pattern Recognition (PR)

- 2 PR and structured-output prediction ▷ 8
3 A running example: Kariotype recognition ▷ 12

Interactive PR (IPR): Feedback and multimodal processing

- 4 Directly benefit from human feedback ▷ 20
5 Non-deterministic feedback and multimodal IPR ▷ 31

User Models, Interaction Protocols and Assessment

- 6 *Passive, active and other interaction protocols* ▷ 41
7 Estimating user interaction effort ▷ 49

Interaction-driven learning

- 8 Adaptive, on-line, active and reinforcement learning ▷ 54

Applications, Final Remarks and Bibliography

- 9 Applications ▷ 61
10 Future work and Conclusions ▷ 65
11 Bibliography ▷ 68

Interaction Protocols and Assessment

In interactive systems, the operator may generally choose many ways or “*interactive actions*” to provide the interaction feedback.

To allow for proper implementations, human creativity has to be limited or predicted in some way so that the system can take maximum advantage of the allowed or expected interactive actions.

In the HCI literature, this kind of limitation or prediction of operator actions is often referred to as *User Model*.

Here only mathematically tractable user models are considered and the set of actions and the ways the user is allowed or expected to make use of these actions is called “*Interaction Protocol*”. A good *Interaction Protocol* must:

- Foster comfortable and productive system-user cooperation
- Permit efficient implementations, since interactive processing is generally highly demanding in terms of response times
- Allow for automated testing & assessment procedures (discussed latter)

The design of a good, friendly, effective and efficient interaction protocol is perhaps the most critical design task for a given IPR application.

General Types of Interaction Protocols

The most basic taxonomy attends to the way it is decided which *hypothesis elements* (HEs) may require human supervision.

Passive: The operator decides which HEs need supervision

Left-to-right: HEs are supervised in fixed order

Desultory: HEs are supervised in unspecified order

Active: The system decides which HEs should be supervised

- A *passive* protocol guarantees “perfect” results from the operator point of view, since her supervision decisions cater for the accurateness of results
- With an *active* protocol the quality of the results depends on the system ability to select appropriate hypothesis elements for supervision

Active interaction allows to trade accuracy for human interaction effort

Example: Interaction Protocols for karyotyping

Passive, left-to-right. Is the protocol that has been assumed in all the examples so far. First, chromosome images are sorted according to their max-posterior probability in order to allow for the greedy search approach. In the successive interaction steps, the operator is assumed to follow this order for supervision

Passive, desultory. The operator might well prefer not to check the partial karyotype correctness in a strict left-to-right order, but perhaps by choosing herself which is “the worst” or most notorious labelling error at each interaction step

Active. At each interaction step, the system computes some *confidence measure* for each chromosome label provided in that step. The one with lowest confidence is proposed for operator supervision. Then the operator validates or corrects *this* label and the system uses the corresponding feedback (and history) to compute its next prediction.

Example: Interactive Karyotyping results

- Same experimental conditions as for classical PR karyotyping results
- Interactive (IPR) results using both greedy and B & B search

Number of corrections needed (%)

Approach	Chromosome	Karyotype
Individual Chromosomes	8.0	76
Using prior $P(h)$ (greedy)	3.7	27
Using prior $P(h)$ (B & B)	2.2	15
Passive IPR (greedy)	2.1	27
Passive IPR (B & B)	1.1	15
Active IPR (B & B)	1.0	15

[Oncina & Vidal, 2011]

Left-to-right Interactive-Predictive Processing

The passive, left-to-right protocol is perhaps the simplest and most appropriate protocol when *output hypotheses* can naturally be structured in terms of *sequences*. It is often referred to as “*left-to-right interactive-predictive*”.

Let h be a sequence of elementary output hypotheses, $h_1, h_2, \dots$ in Eq. (8):

$$\hat{h} = \arg \max_{h \in \mathcal{H}} P(h \mid x, h', d)$$

The history h' and the (deterministic) corrective feedback d can be jointly considered as a *correct prefix*, p , of h , leading to:

$$\hat{h} = \arg \max_{h \in \mathcal{H}} P(h \mid x, p) = \arg \max_{h \in \mathcal{H}} P(x \mid p, h) P(h \mid p) \quad (24)$$

$P(h \mid p)$ should be null for those h that do not have p as a prefix, which implies that $\hat{h}$ must be the concatenation of the given p and some optimal *suffix* $\hat{s} \in \mathcal{H}'$, the set of possible suffixes. Then, Eq. (24) can be written as:

$$\hat{s} = \arg \max_{s \in \mathcal{H}'} P(s \mid x, p) = \arg \max_{s \in \mathcal{H}'} P(x \mid p, s) P(s \mid p) \quad (25)$$

Interaction with Weaker Feedback

In many cases, the operator may like to just point the place where an error exists and wait for the system to change its hypothesis, trying to anticipate the correction which she has in mind. This simple user action is often called “click”.

In equation (9), let d be just the *index* of the wrong hypothesis element:

$$\hat{h} = \arg \max_{h \in \mathcal{H}} P(x | h) P(h | h', d)$$

where

$$P(h | h', d) \propto \begin{cases} 0 & \text{if } h_d = h'_d \\ P(h | h') & \text{otherwise} \end{cases} \quad (26)$$

and $P(h | h')$ accounts for the prior probability of a hypothesis, conditioned only by the (uncorrected) history, h' .

Since “click” actions are often used repeatedly, the successive values of h'_d must be cached and $P(h | h', d)$ must be computed taking into account all the previously discarded values of h'_d (not just the one from the previous step).

Interaction without Input Data

There are interactive applications in which no input data, x , is given.

An example is the interactive generation of text, where an IPR system assists the user for writing text by predicting what are the most probable continuations of the text produced so far.

Other applications, such as Interactive Music Composition and Relevance-based Image Retrieval, can also be considered in this category.

In these cases the formulation is essentially a trivial simplification of Eq. (8):

$$\hat{h} = \arg \max_{h \in \mathcal{H}} P(h | x, h', d) = \arg \max_{h \in \mathcal{H}} P(h | h', d) \quad (27)$$

If the protocol is Left-t-Right Interactive-Predictive, the problem reduces to predict a best suffix $\hat{s}$, given a known prefix p . From Eq. (25):

$$\hat{s} = \arg \max_{s \in \mathcal{H}'} P(s | p) \quad (28)$$

where $\mathcal{H}'$ is the set of possible suffixes.

Index

Introduction

- 1 Notation and introduction to Interactive Pattern Recognition ▷ 0

Classical Pattern Recognition (PR)

- 2 PR and structured-output prediction ▷ 8
- 3 A running example: Kariotype recognition ▷ 12

Interactive PR (IPR): Feedback and multimodal processing

- 4 Directly benefit from human feedback ▷ 20
- 5 Non-deterministic feedback and multimodal IPR ▷ 31

User Models, Interaction Protocols and Assessment

- 6 Passive, active and other interaction protocols ▷ 41
- 7 *Estimating user interaction effort* ▷ 49

Interaction-driven learning

- 8 Adaptive, on-line, active and reinforcement learning ▷ 54

Applications, Final Remarks and Bibliography

- 9 Applications ▷ 61
- 10 Future work and Conclusions ▷ 65
- 11 Bibliography ▷ 68

Assessing IPR systems

- The definition of an interaction protocol has strong implications in system testing
- Testing with a real operator working with the system is too expensive for day-to-day system development work
- “Objective” assessment procedures are needed which can be based on *labelled testing corpora*, as in the time-honored tradition of classical PR
 - This requires an unambiguously defined interaction protocol
 - But not every interaction protocol lends itself to corpus-based assessment. This adds to the set of tradeoffs to consider in the development of IPR systems
- Decision theory provides an adequate framework to rigorously define assessment criteria in terms of loss functions. But, again, not every loss function leads to mathematically tractable decision functions

User effort estimation

- In the IPR framework performance has to be gauged mainly in terms of how much human effort is required to achieve the goals of the considered task
- This requires human work and judgement, but by precisely specifying goals and ground-truth, corpus-based testing is still applicable in most IPR tasks
 - A testing corpus for *traditional, non-interactive* PR typically consists of a collection of objects, accompanied by their correct (structured) labellings
Assessment consists in counting elementary hypothesis errors (i.e., number of times a system hypothesis element differs from the correct label)
 - For many interaction protocols similar corpora and labelling can be used for assessing *interactive performance* in terms of *estimated user effort*
- In IPR, we should not focus on errors (the operator ensures the required accuracy), but reference labellings can be used to determine how many interaction steps are needed to produce a fully correct hypothesis
- For many interaction protocols, *user effort estimates* can be easily obtained from counts of required interaction steps

Example: interaction effort estimation in karyotyping

- The protocol considered in interactive karyotyping was *left-to-right*
- User interaction effort was estimated in terms of the number of user corrective interactions needed to produce correct labelling. This was made automatically using a *reference test-set* labelling:
 - At each interaction step, user behaviour is simulated by computing the *longest common prefix*, p' between the current system hypothesis and the corresponding reference labelling
 - Then the first system wrong hypothesis element after this common prefix is replaced with the correct reference label, r , and the number of corrective interactions is increased by one
 - Finally, the resulting correct prefix, $p = p'r$, is used by the IPR system to compute a new suffix prediction, $\hat{s}$, as in Eq. (25)
- This testing paradigm (adequately) ignores user *supervision* effort; that is, only *corrective* interaction steps are considered relevant in order to measure (estimate) system/user performance

IPR assessment: final remarks

- In general, IPR performance measures should take into account (perhaps with different costs) both *corrective* and *supervision* interaction steps
- Measuring only *corrective* steps may be adequate in *passive interaction*, often used to guarantee perfect results:
 - in this case a complete supervision of all the system hypotheses is required and only corrective effort may make a difference in performance
- When an IPR system is considered sufficiently mature, final testing should be based on evaluations with human operators actually working with the real tasks the system is designed for. However, this kind of evaluation:
 - is too subjective to be useful to guide early development decisions
 - is too expensive and time consuming to be carried out frequently
 - is affected by many factors which are far away from the fundamental principles upon which system design is based
- How the final *User Interface* (UI) is designed is one of these important factors. A good design should take into account the IPR design principles and, in particular, the assumed *interaction protocol*

Index

Introduction

- 1 Notation and introduction to Interactive Pattern Recognition ▷ 0

Classical Pattern Recognition (PR)

- 2 PR and structured-output prediction ▷ 8
- 3 A running example: Kariotype recognition ▷ 12

Interactive PR (IPR): Feedback and multimodal processing

- 4 Directly benefit from human feedback ▷ 20
- 5 Non-deterministic feedback and multimodal IPR ▷ 31

User Models, Interaction Protocols and Assessment

- 6 Passive, active and other interaction protocols ▷ 41
- 7 Estimating user interaction effort ▷ 49

Interaction-driven learning

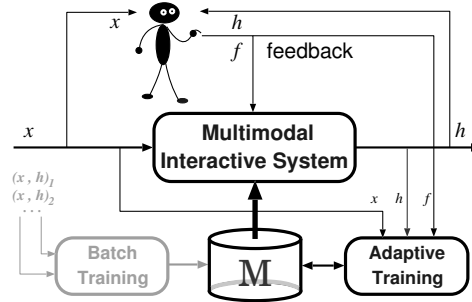
- 8 *Adaptive, on-line, active and reinforcement learning* ▷ 54

Applications, Final Remarks and Bibliography

- 9 Applications ▷ 61
- 10 Future work and Conclusions ▷ 65
- 11 Bibliography ▷ 68

Interaction-driven learning

So far all models, $\mathcal{M}$, needed for IPR have been assumed to be fixed. But now *human interaction* offers another unique opportunity to improve system's behaviour by tuning the models, $\mathcal{M}$. The feedback produced at each step of the interaction process can generally be converted into new, fresh training information, useful for *adapting* the system to changing environment.



For many years, *adaptive learning* and other related learning paradigms such as *on-line*, *semi-supervised*, *reinforcement*, *active*, etc.) have been the focus of thorough studies. However, most of these studies are mainly theoretically oriented. Practical applications of the theoretical results are generally scarce, mainly because only the *interactive* paradigm offers a natural framework where these learning paradigms can be used advantageously. The application of these ideas in our IPR framework require establishing adequate training criteria. These criteria should allow the development of adaptive training algorithms that take the maximum advantage of the interaction-derived data to ultimately minimise the overall human effort in the long term.

IPR and Online Learning (OL)

In IPR, the models $\mathcal{M}$ are initially trained with a batch, *seed* corpus $T = \{(x, h)_i\}$, as in traditional PR. In successive interaction steps, the system gathers new correct input-output pairs $T' = \{(x', h')_j\}$

Simple OL idea: train $\mathcal{M}$ by merging both data sets T and T' .

- Also called *incremental learning*, since T' is seen as an “increment” to T
- Efficient whenever learning can rely on updating *sufficient statistics*
 - Just update event counts for simple models (e.g., Gaussian, N-grams ...)
 - Require *Incremental Expectation–Maximisation* (EM) for models with hidden (latent) variables [Neal & Hinton, 1998]
- Need a trade-off between the impact of T and T' (T_k , $k = 1, 2, \dots$, in general):

- Linear interpolation:

$$P_{\alpha}(h \mid \dots) = \sum_{k=1}^K \alpha_k \cdot P_{\mathcal{M}_k}(h \mid \dots), \quad \sum_{k=1}^K \alpha_k = 1 \quad (29)$$

- Log-linear modelling:

$$P_{\lambda}(h \mid \dots) = \frac{1}{Z_{\lambda}} \exp\left(\sum_{k=1}^K \lambda_k \log P_{\mathcal{M}_k}(h \mid \dots)\right) \quad (30)$$

- Bayesian approaches

IPR and Active Learning (AL)

A set of *unsupervised* training samples, T , is given. AL techniques automatically select, from T , a minimum set of samples, T' , to be (manually) supervised or labelled. Training with T' should lead to best system performance [Dasgupta, 2009, Hanneke, 2009].

- AL techniques address the “*sampling bias*” problem; i.e., distortion in the sample probability distribution, with respect to the natural distribution, produced by the AL sampling strategies.
- AL is particularly useful for *Active Interaction* protocols: Selecting good hypothesis elements to be supervised should serve to improve both prediction and training
- The tandem *AL + Active Interaction* enables useful trade-offs between overall accuracy and interaction effort (supervision + correction)
- *Semisupervised training* techniques can be useful to improve training by using samples which have not been selected for supervision

IPR and Reinforcement Learning (RL)

In interaction protocols based on *weak feedback*, the feedback given by the user is generally not totally informative. This is directly related with *learning with limited-feedback* [Shalev, 2008], a branch of RL [Auer, 2008].

Also relevant to model user’s preferences; e.g., to select, among the available *interactive actions*, those most promising for best (active) IPR performance.

- An RL system tries to maximise the “*benefit*” it can obtain from the environment, using two confronted strategies: *exploration* and *exploitation*
- This is formalised in terms of minimizing the “*regret*”; i.e., is the difference between the actual benefit and the maximum benefit that could be obtained
- Let $B(h^{(1)}, \dots, h^{(T)})$ be the benefit (e.g, accuracy) obtained from the last T hypotheses, $h^{(1)}, \dots, h^{(T)}$. Then the regret is:

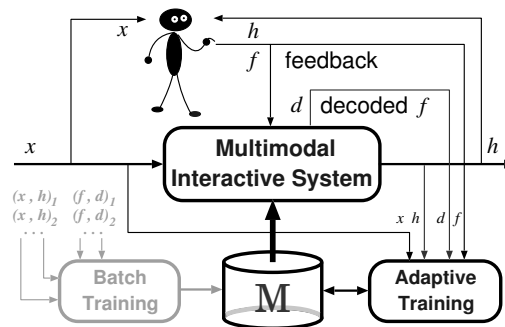
$$R(h^{(1)}, \dots, h^{(T)}) = \max_{h'^{(1)}, \dots, h'^{(T)}} B(h'^{(1)}, \dots, h'^{(T)}) - B(h^{(1)}, \dots, h^{(T)}) \quad (31)$$

- RL uses Dynamic Programming to obtain an (exploration-exploitation) *optimal policy* to minimize R by selecting appropriate actions at each step

Non-deterministic feedback decoding and Online Learning

The concept of *Adaptive Learning* using interactively produced training data applies not only to the main system models (needed to obtain $\hat{h}$ for given x), but also to the Models needed for feedback decoding.

The data needed for this adaptation is directly available from the explicit feedback decoding given by the solution of (19), or its approximations (20-23).



$\mathcal{M}$ includes models for both *main* and *feedback* data processing. Both are initially trained in batch mode and then successively *adapted* to the task and/or the user by using training pairs derived from the user feedback information.

Example: Adapting e-pen feedback models for karyotyping

- The HTR likelihood (HMM) models, $P(t | l)$ for feedback decoding can be easily adapted to the specific handwriting style of the user.

The required training data are pairs (t, l) , where t is an *e-pen* trajectory and l is the correct text associated with t (a label from “1” to “22”).

These pairs become readily available after every successful corrective interaction step.

- The feedback decoding (conditioned) prior, $P(l | \dots)$, can be easily adapted to the typical errors made by the IPR chromosome recognizer: just tune label priors according to the observed label error frequencies.

This adaptation requires just label error counts, an information which is also readily available after each successful interaction step.

Index

Introduction

- 1 Notation and introduction to Interactive Pattern Recognition ▷ 0

Classical Pattern Recognition (PR)

- 2 PR and structured-output prediction ▷ 8
- 3 A running example: Kariotype recognition ▷ 12

Interactive PR (IPR): Feedback and multimodal processing

- 4 Directly benefit from human feedback ▷ 20
- 5 Non-deterministic feedback and multimodal IPR ▷ 31

User Models, Interaction Protocols and Assessment

- 6 Passive, active and other interaction protocols ▷ 41
- 7 Estimating user interaction effort ▷ 49

Interaction-driven learning

- 8 Adaptive, on-line, active and reinforcement learning ▷ 54

Applications, Final Remarks and Bibliography

- 9 *Applications* ▷ 61
- 10 Future work and Conclusions ▷ 65
- 11 Bibliography ▷ 68

MIPR Applications

- Computer Assisted Transcription:
Text Images (CATTI), Speech (CAST) and Music
- Multimodal Interaction for Document Analysis
- Interactive Machine Translation (IMT)
- Interactive Text Generation and Music Composition
- Relevance-based Information Retrieval
- Multimodal Interactive Image and Video processing
- . . .
- Many other possible applications; see:

<http://miprcv.prhlt.upv.es>

The MIPRCV Research Programme (2007-2012)

Multimodal Interacción in Pattern Recognition and Computer Vision (MIPRCV)

(5-year programme, 7 research groups, 90+ PhD researchers)



<http://miprcv.prhlt.upv.es>

Objectives: Explore the challenges and oportunities of MI in PR & CV

MIPRCV: Technologies & Applications

Iram pande mihi Pelidae, Diua,
Tristia quae miseris iniecit fun
Atque animas fortes heroum tr
Latrantumque **dedit** | **ἐ** ἐλώρια
οἰωνοῖσι τε πᾶσι· Διὸς δ' ἐτέλει
ἐξ οὗ δὴ τὰ πρῶτα διαστήτην ἐξ
Ἀτρειδῆς τε ἄναξ ἀνδρῶν καὶ δῖ

Index

Introduction

- 1 Notation and introduction to Interactive Pattern Recognition ▷ 0

Classical Pattern Recognition (PR)

- 2 PR and structured-output prediction ▷ 8
3 A running example: Kariotype recognition ▷ 12

Interactive PR (IPR): Feedback and multimodal processing

- 4 Directly benefit from human feedback ▷ 20
5 Non-deterministic feedback and multimodal IPR ▷ 31

User Models, Interaction Protocols and Assessment

- 6 Passive, active and other interaction protocols ▷ 41
7 Estimating user interaction effort ▷ 49

Interaction-driven learning

- 8 Adaptive, on-line, active and reinforcement learning ▷ 54

Applications, Final Remarks and Bibliography

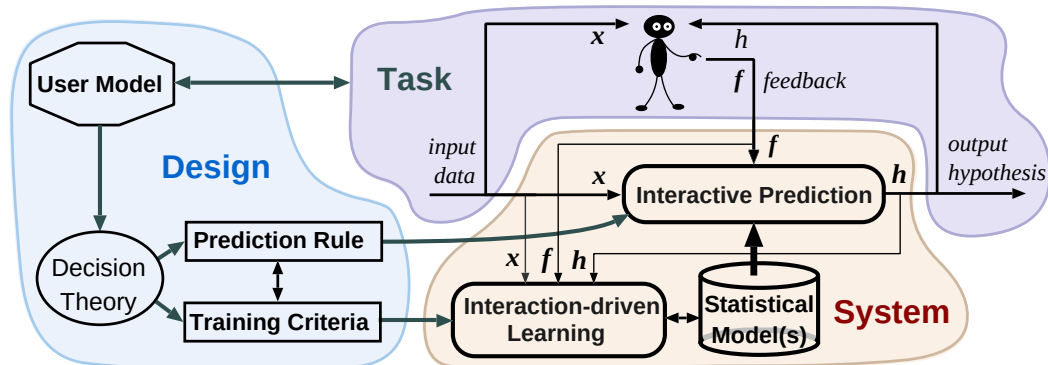
- 9 Applications ▷ 61
◦ 10 *Future work and Conclusions* ▷ 65
11 Bibliography ▷ 68

Future work: Decision Theory and IPR

Inter-related aspects of IPR development:

- Design of user modeling and interaction protocols
- Develop interactive prediction algorithms
- Develop interaction-driven learning approaches

An adequate common, integrating framework: *Decision Theory*



Concluding remarks: future of intelligent systems?

Fully autonomous artificial systems with human-like intelligence:

- Fallacious ambition of humanity
- How far are we from really knowing how to do it? decades? centuries? millenia?
- Do we really need, want, or like it?

Interactive, computer-assisted perception & cognition:

- Assist persons in useful tasks that require non-trivial perceptive/cognitive skills
- Amplify human “intelligence”
- Maybe it is less ambitious than “full automation”, but it is:
 - Realistic,
 - Possible: we know or are close to know how to do it properly,
 - We do need, want and like it

Multimodal Interaction in Pattern Recognition:

- Interesting research *challenges and opportunities* in many applications where technology is expected to *assist*, rather than *replace* the human agents

Index

Introduction

- 1 Notation and introduction to Interactive Pattern Recognition ▷ 0

Classical Pattern Recognition (PR)

- 2 PR and structured-output prediction ▷ 8
- 3 A running example: Karyotype recognition ▷ 12

Interactive PR (IPR): Feedback and multimodal processing

- 4 Directly benefit from human feedback ▷ 20
- 5 Non-deterministic feedback and multimodal IPR ▷ 31

User Models, Interaction Protocols and Assessment

- 6 Passive, active and other interaction protocols ▷ 41
- 7 Estimating user interaction effort ▷ 49

Interaction-driven learning

- 8 Adaptive, on-line, active and reinforcement learning ▷ 54

Applications, Final Remarks and Bibliography

- 9 Applications ▷ 61
- 10 Future work and Conclusions ▷ 65
- 11 *Bibliography* ▷ 68

Bibliography

- R.Neal, G.E.Hinton. "A view of the em algorithm that justifies incremental, sparse, and other variants". In Learning in Graphical Models, pp.355-368. Kluwer Academic Pub. 1998.
- E. Vidal, F. Casacuberta, L. Rodríguez, J. Civera and C. Martínez. "Computer-assisted translation using speech recognition". IEEE Trans. on Audio, Speech and Language Proc, 14(3):941-951, 2006.
- L. Rodríguez, F. Casacuberta, and E. Vidal. "Computer Assisted Transcription of Speech" Proc. of the Iberian Conf. on Pattern Recognition and Image Analysis, Vol.4477 of LNCS, pp.241-248, 2007.
- E. Vidal, L. Rodríguez, F. Casacuberta and I. García-Varea: "Interactive Pattern Recognition". 4th Joint Workshop on Multimodal Interaction and Related Machine Learning Algorithms (MLMI-07), Volume 4892 of LNCS, pp.60-71. 2007.
- S.Shalev-shwartz, A.Tewari. "Efficient bandit algorithms for online multiclass prediction". In Proc. of the 25th Int. Conf. Machine Learning. 2008.
- P.Auer, T.Jaksch, R.Ortner. "Near-optimal regret bounds for reinforcement learning" Tech. Rep, Univ. of Leoben, 2009.
- S.Dasgupta. "The two faces of active learning". DS'09 Proc. of Int. Conf. on Discovery Science, pp.35, Springer, 2009.
- S.Hanneke. "Theoretical foundations of active learning". PhD thesis, CMU-ML-09-106. 2009.
- S.Barrachina, O.Bender, F.Casacuberta, J.Civera, E.Cubel, S.Khadiji, A.Lagarda H.Ney, J.Tomás, E.Vidal. "Statistical approaches to computer-assisted translation". Computational Linguistics, Vol.35(1) pp.3-28, 2009.
- F.Casacuberta, J.Civera, E.Cubel, A.L.Lagarda, G.Lapalme, E.Macklovitch, E.Vidal. "Human interaction for high quality machine translation". Comm. of the ACM, Vol.52(10), pp.135-138, 2009.
- A.H. Toselli, V. Romero, M. Pastor and E. Vidal. "Multimodal interactive transcription of text images". Pattern Recognition, Vol.43, N.5, pp.1814–1825, 2010.
- J.Oncina, E.Vidal: "Interactive Structured Output Prediction: Application to Chromosome Classification". In: Proc. of IbPRIA-2011, Pattern Recognition and Image Analysis (LNCS). Vol. 6669. pp. 256-264. 2011.
- A.H.Toselli, E.Vidal, F.Casacuberta: "Multimodal Interactive Pattern Recognition and Applications". Springer Verlag, 2011.
- V.Romero, A.H.Toselli, E.Vidal: "Multimodal Interactive Transcription of Handwritten Text Images". World Scientific, 2012.